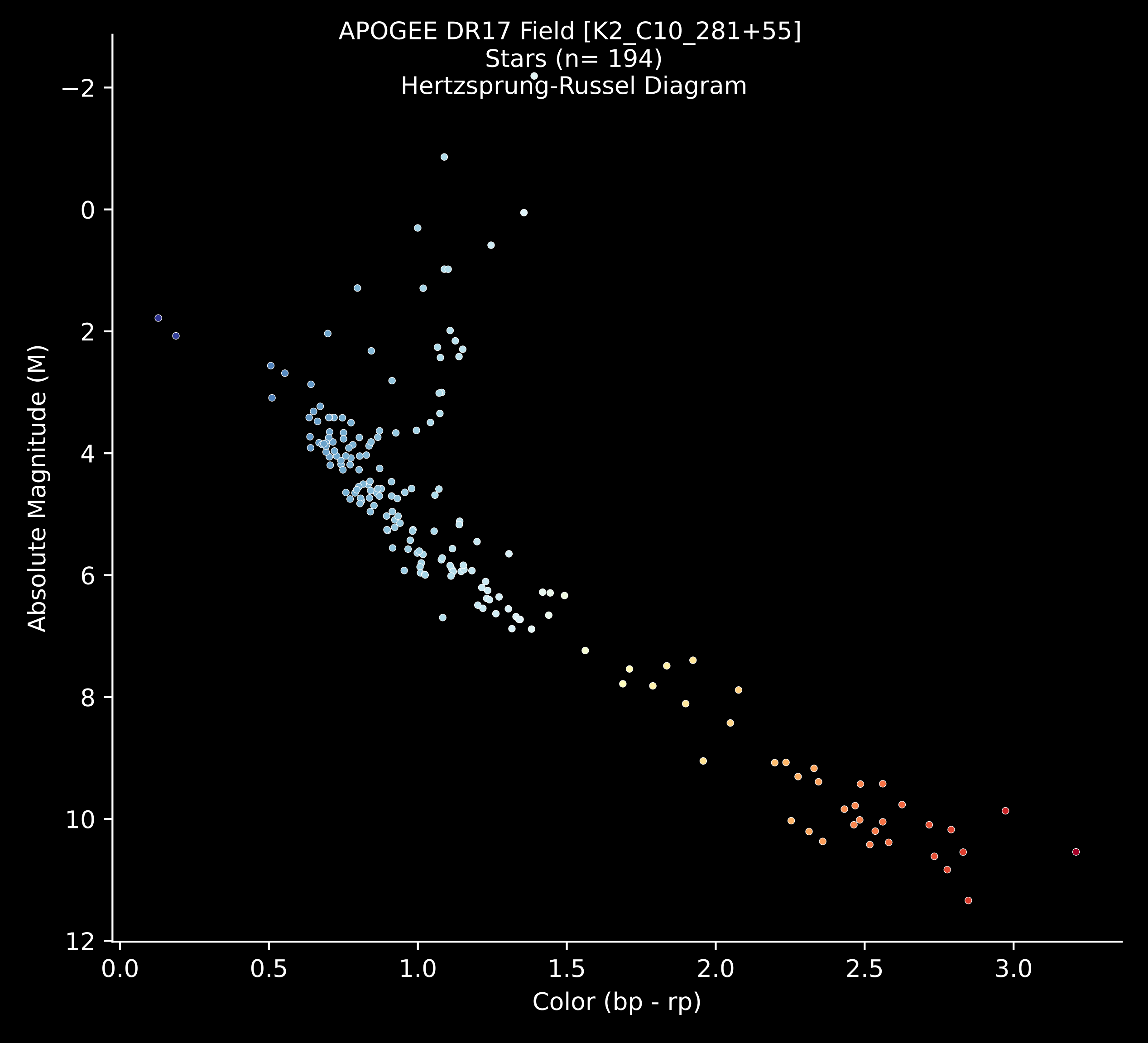
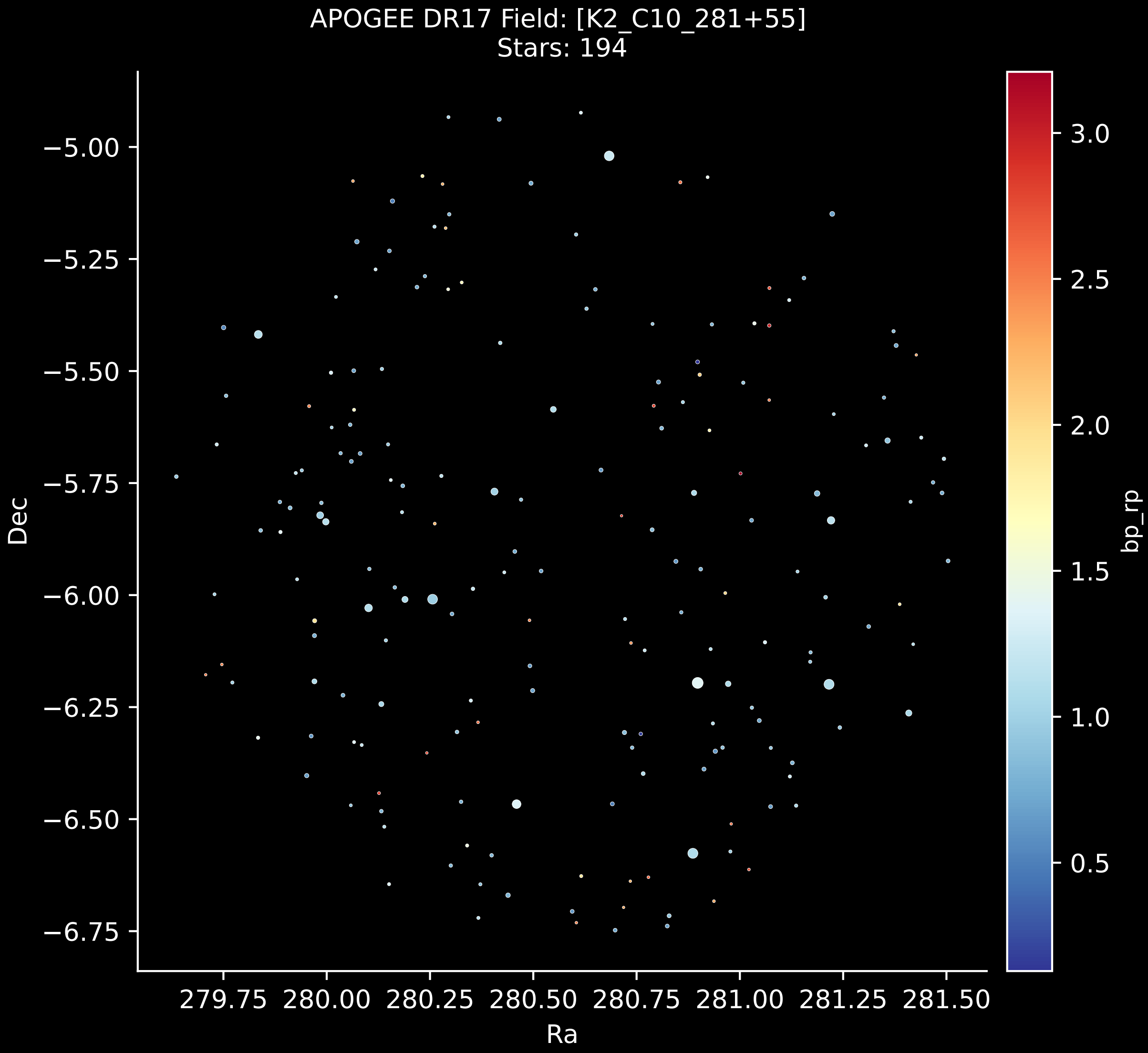
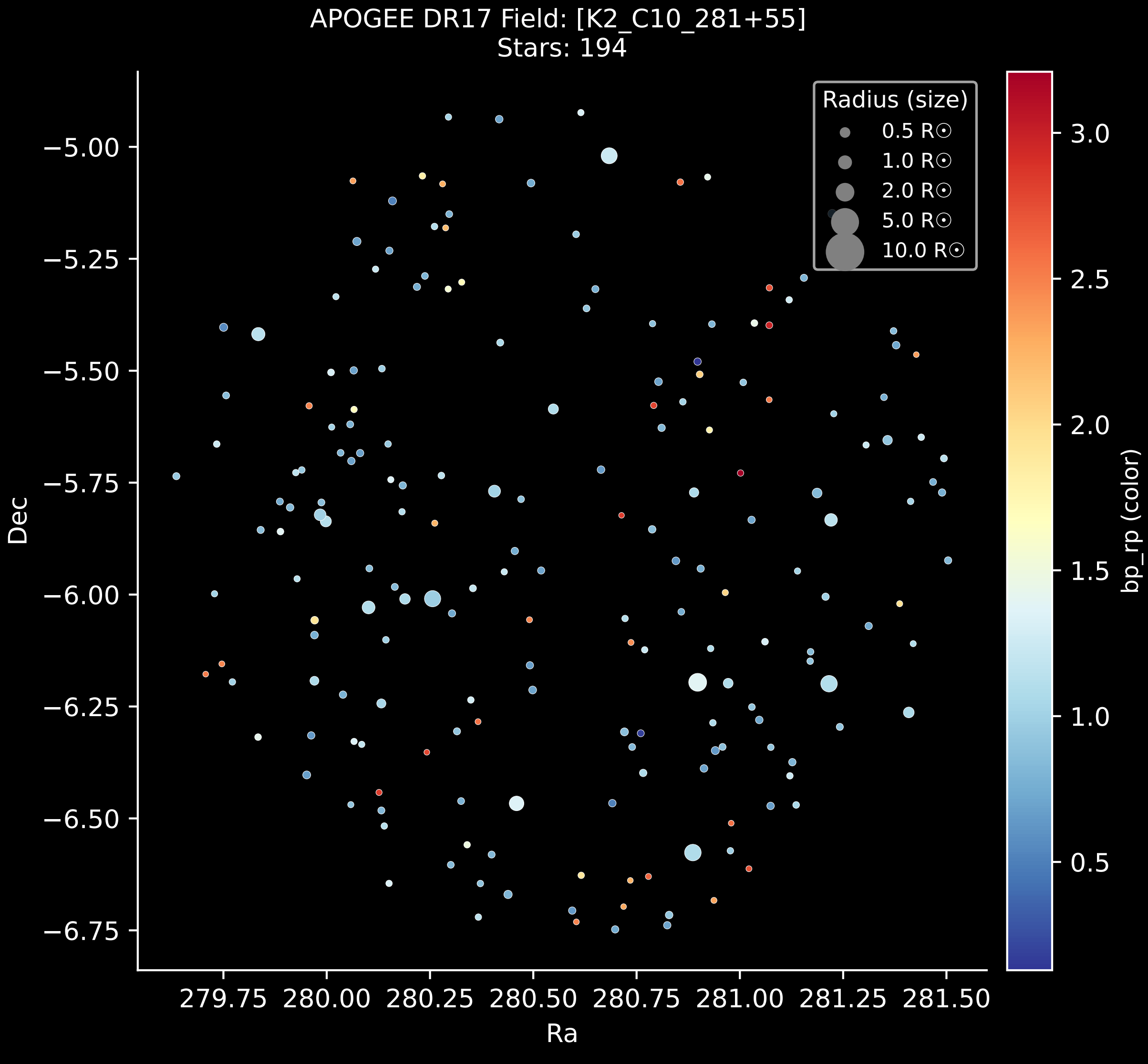
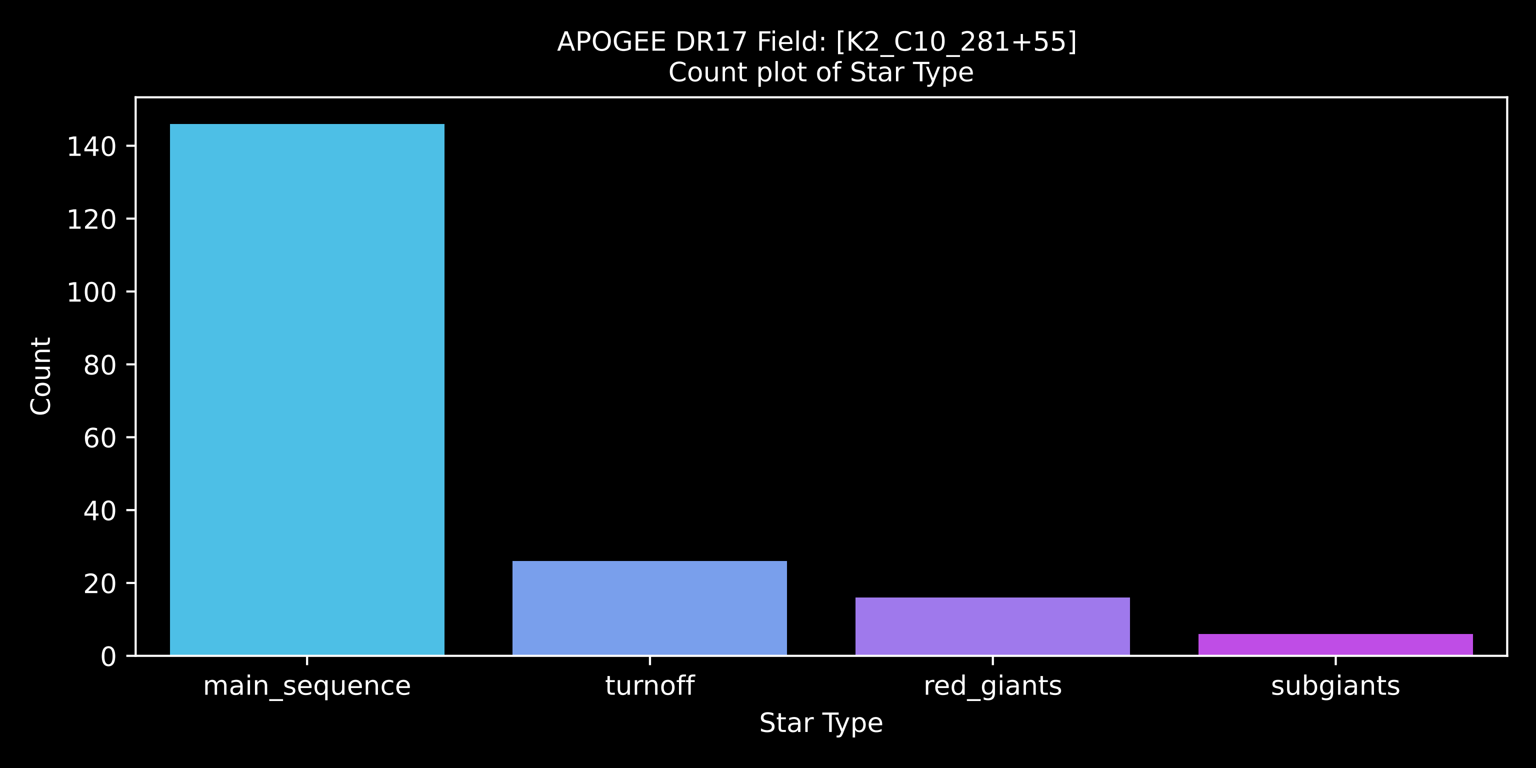
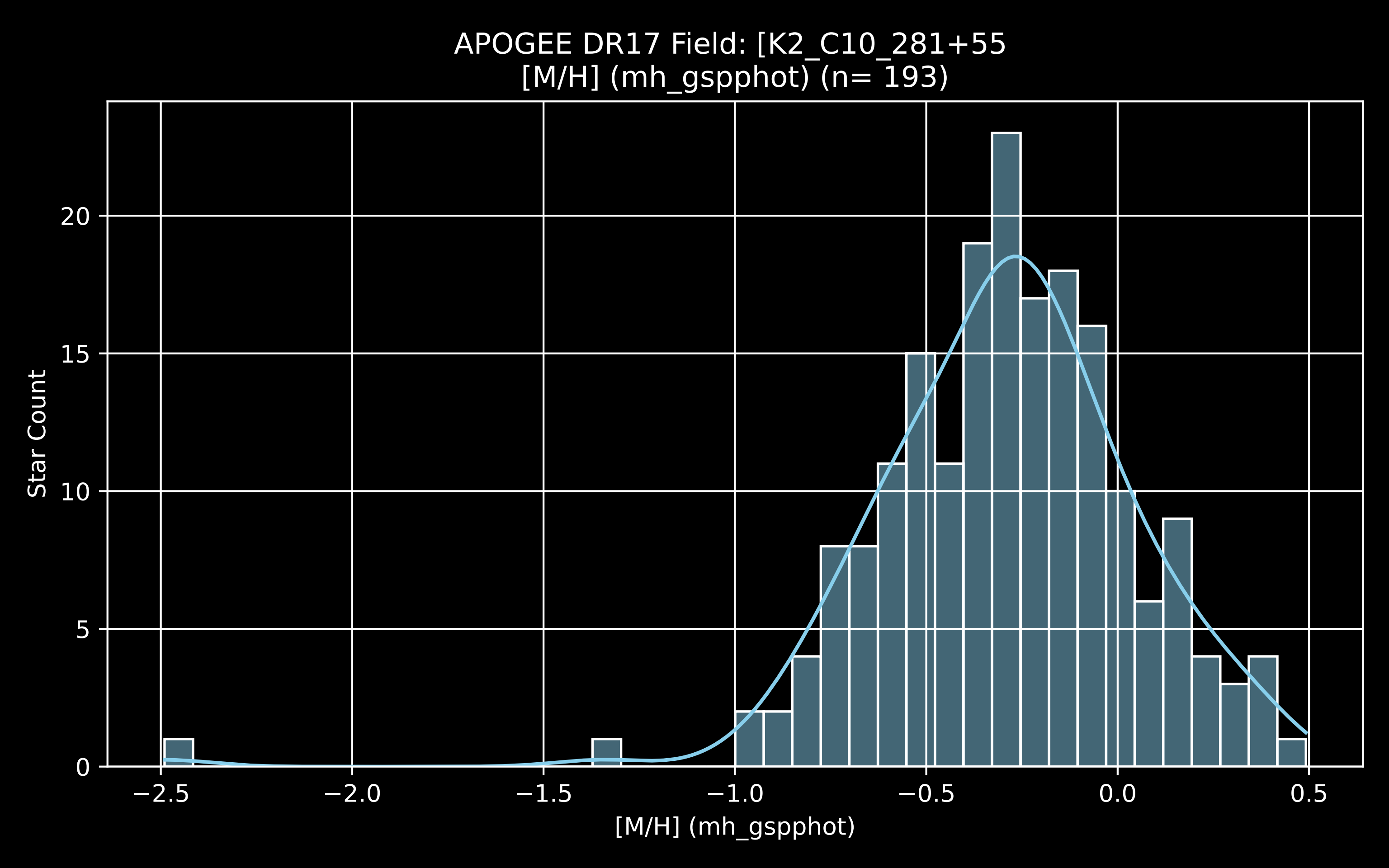

APOGEE DR17 Field [K2_C10_281+55]
Stars (n= 194)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−2
0
2
4
6
8
10
12
0.0
0.5
1.0
1.5
2.0
2.5
3.0

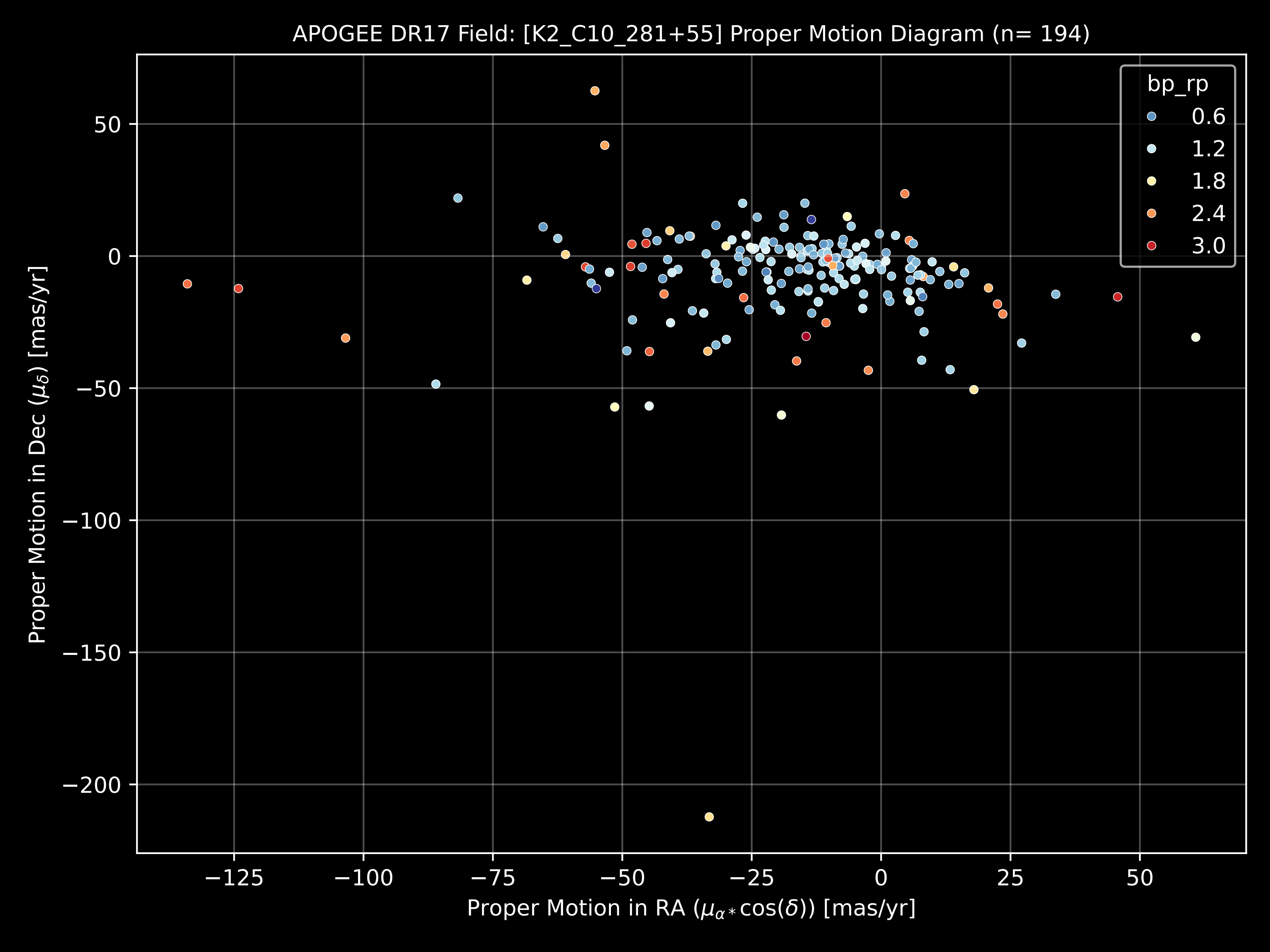

APOGEE DR17 Field: [K2_C10_281+55]
Stars: 194
Dec
Ra
bp_rp
−5.00
−5.25
−5.50
−5.75
−6.00
−6.25
−6.50
−6.75
279.75
280.00
280.25
280.50
280.75
281.00
281.25
281.50
3.0
2.5
2.0
1.5
1.0
0.5

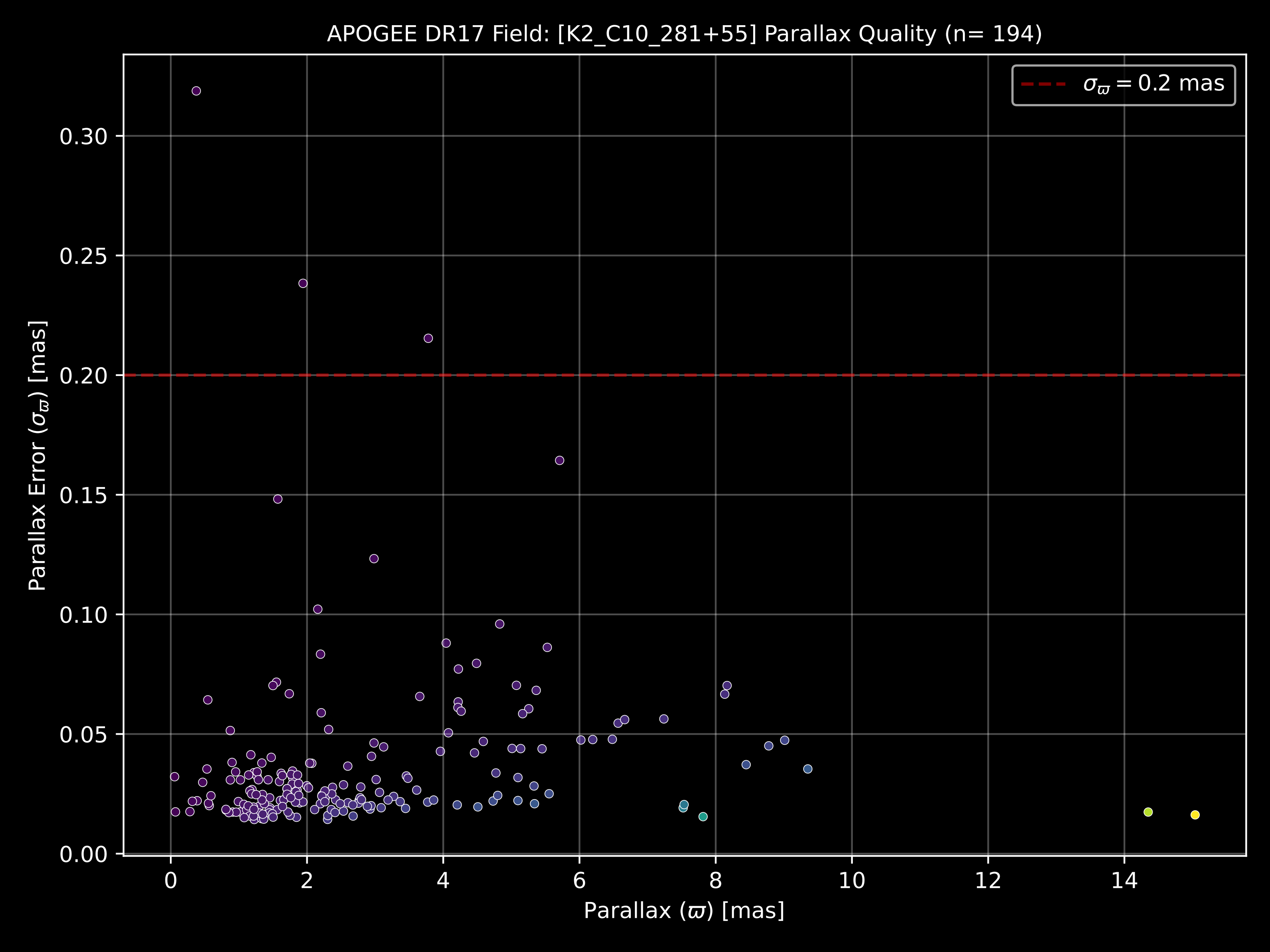

APOGEE DR17 Field: [K2_C10_281+55]
Stars: 194
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
−5.00
−5.25
−5.50
−5.75
−6.00
−6.25
−6.50
−6.75
Dec
279.75
280.00
280.25
280.50
280.75
281.00
281.25
281.50
Ra
3.0
2.5
2.0
1.5
1.0
0.5
bp_rp (color)

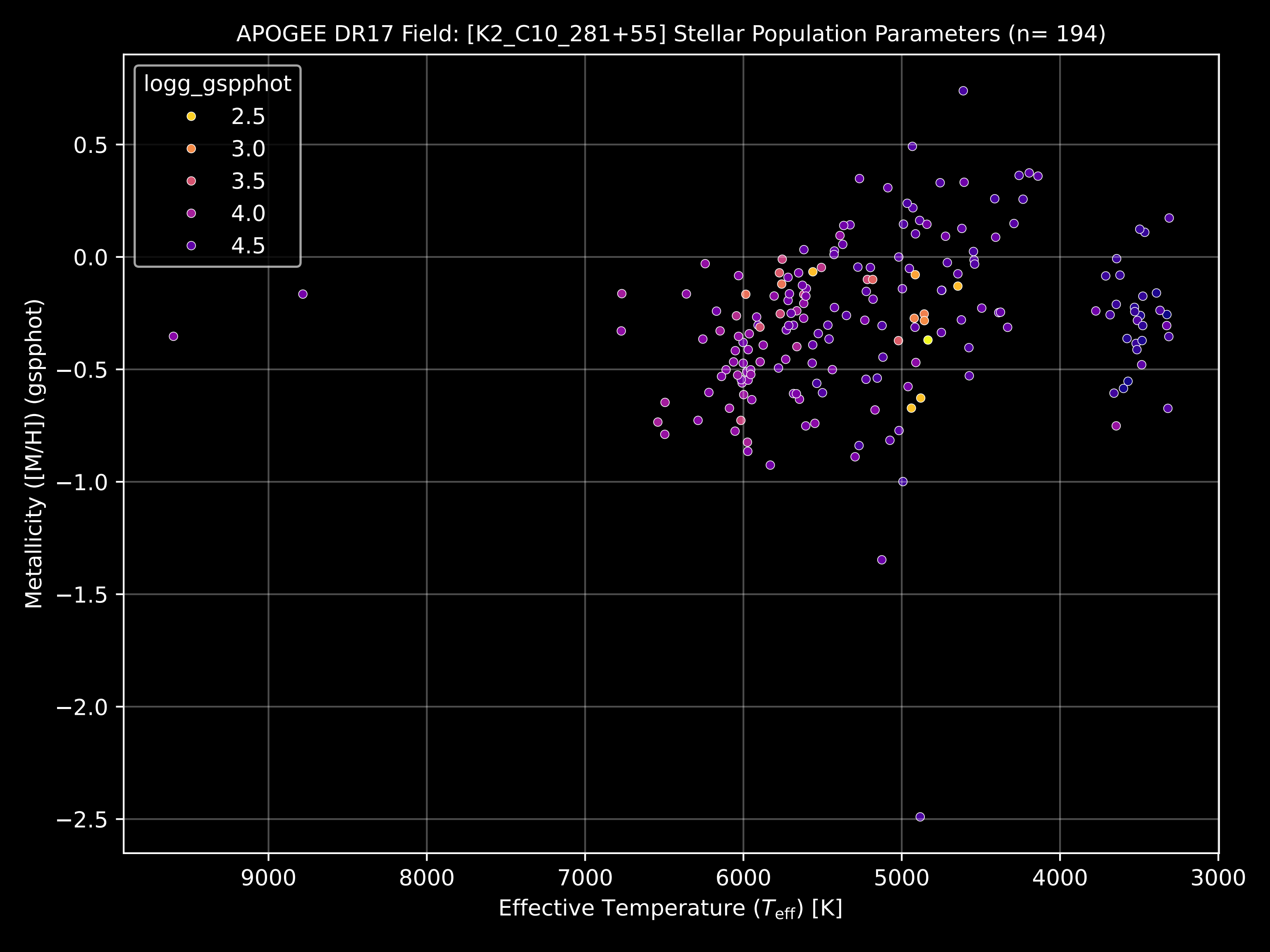

APOGEE DR17 Field: [K2_C10_281+55]
Count plot of Star Type
Count
0
20
40
60
80
100
120
140
main_sequence
turnoff
red_giants
subgiants
Star Type

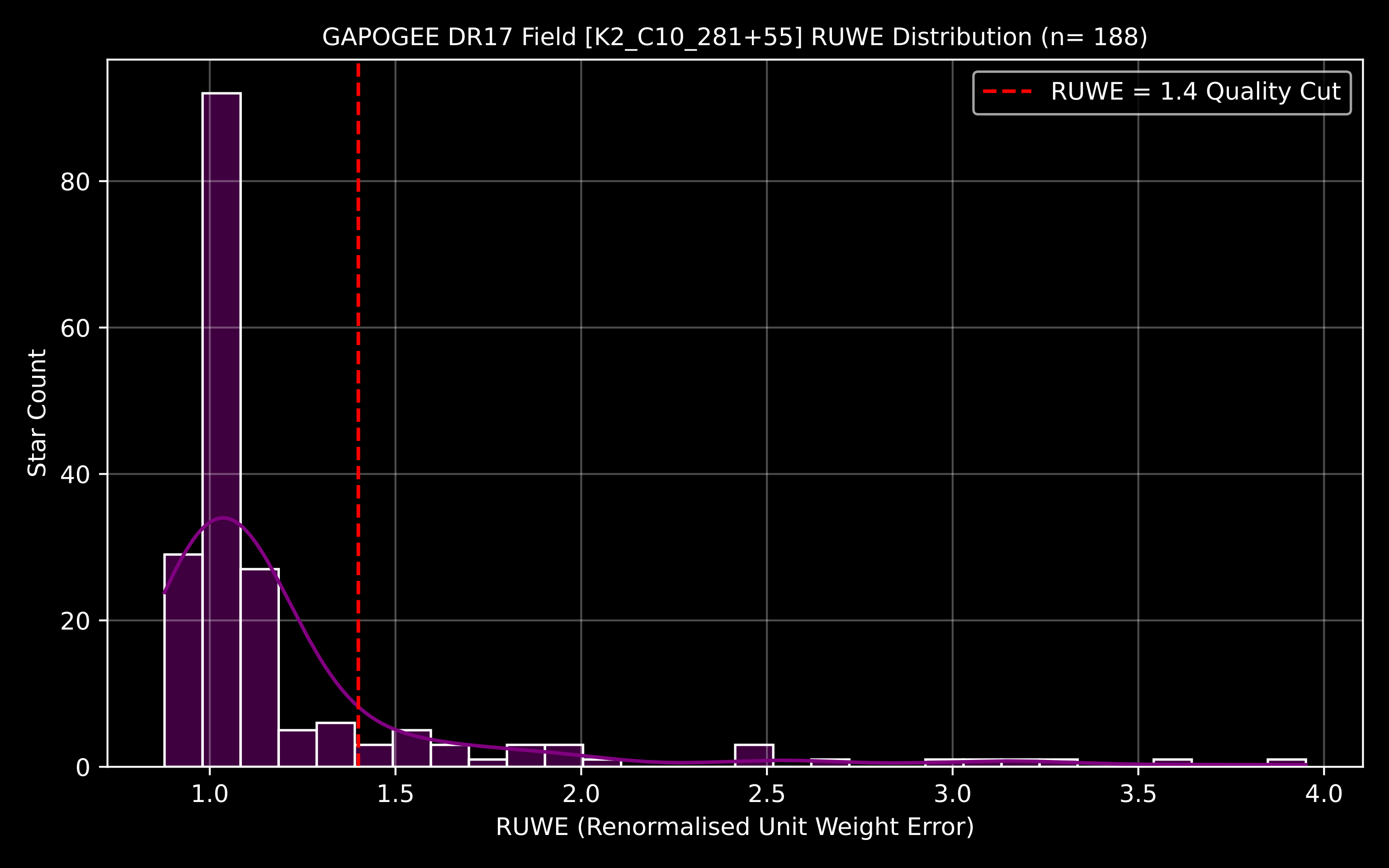


APOGEE DR17 Field: [K2_C10_281+55
[M/H] (mh_gspphot) (n= 193)
Star Count
[M/H] (mh_gspphot)
0
5
10
15
20
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5

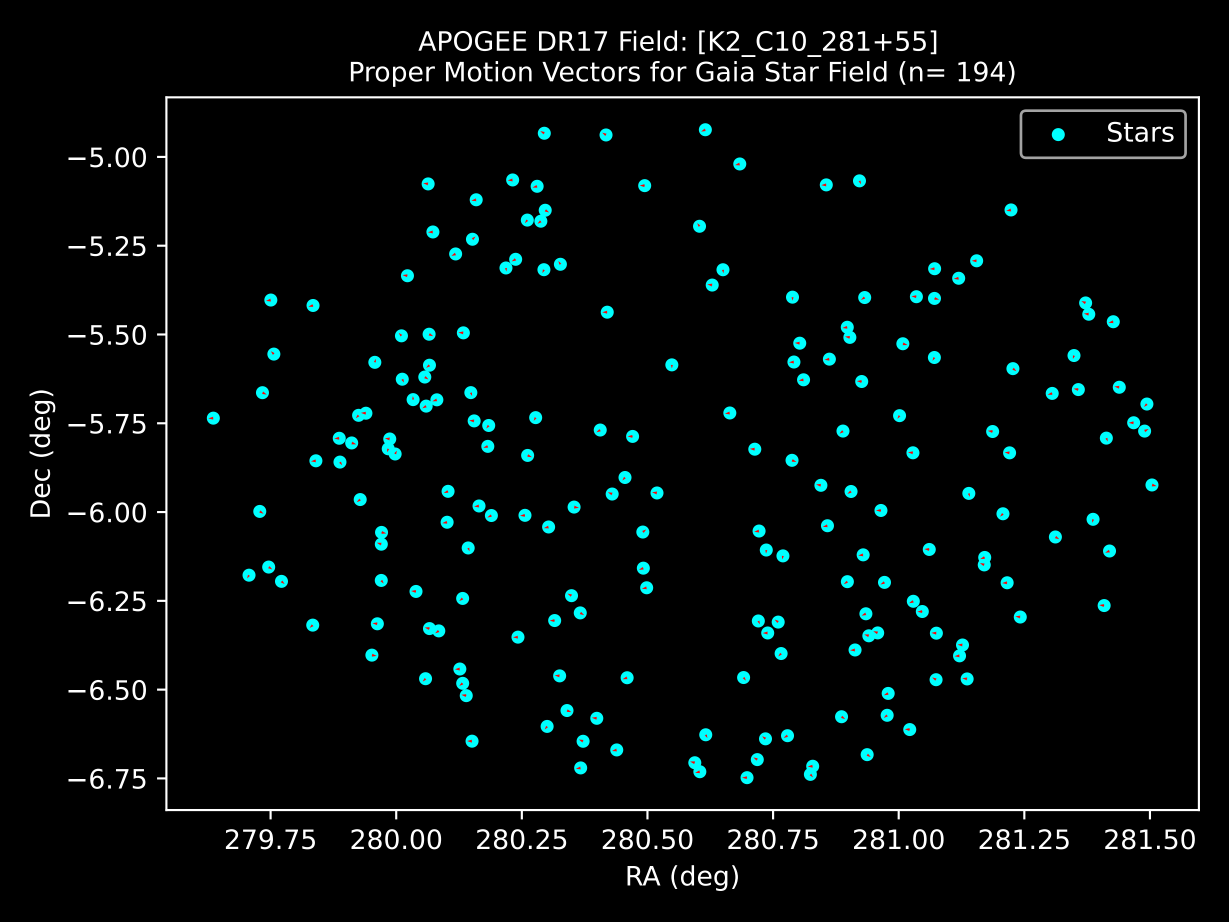

APOGEE DR17 Field: [K2_C10_281+55] Proper Motion Diagram (n= 194)
bp_rp
0.6
1.2
1.8
2.4
3.0
50
0
−50
−100
−150
−200
Proper Motion in Dec ($\mu_\delta$) [mas/yr]
−125
−100
−75
−50
−25
0
25
50
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

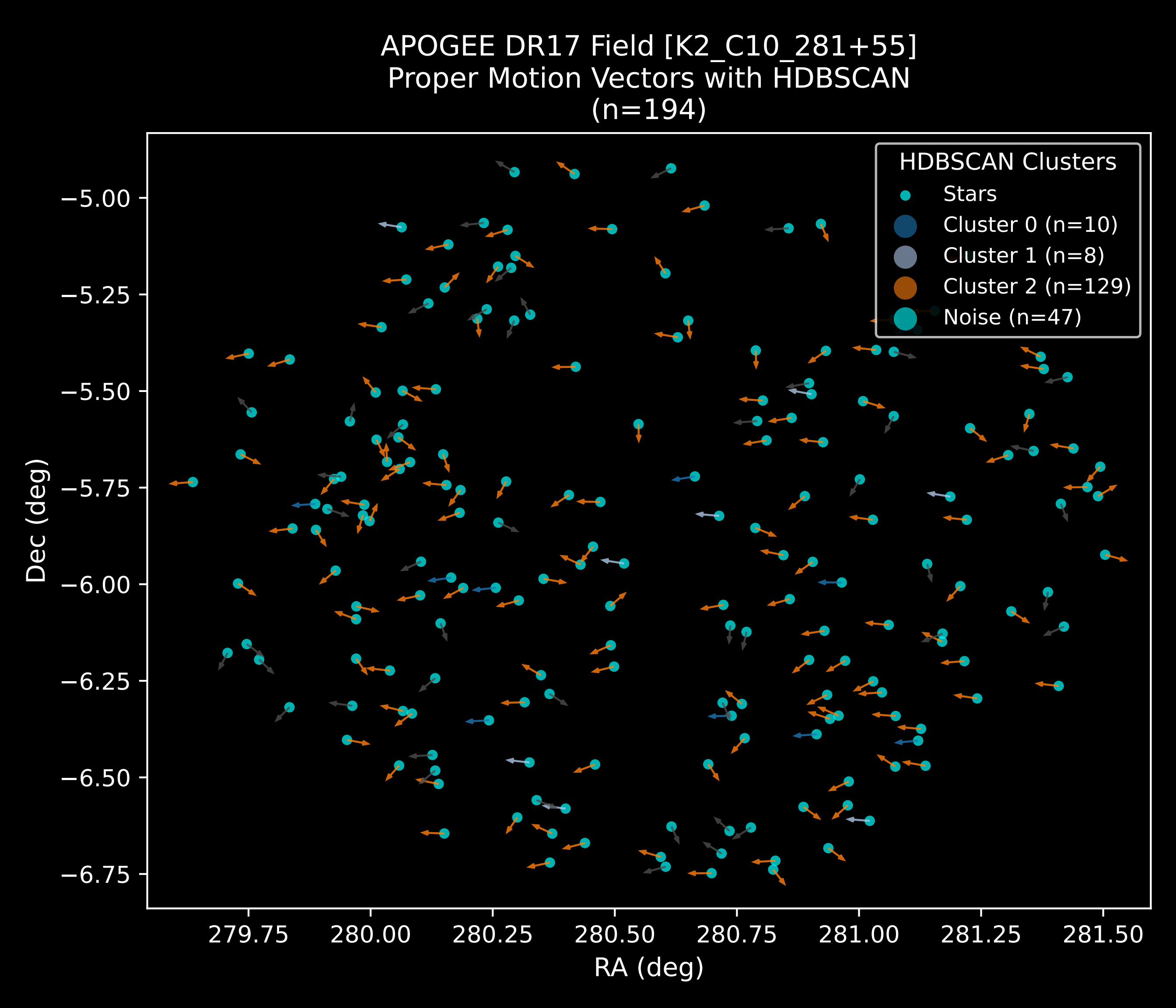

APOGEE DR17 Field: [K2_C10_281+55] Parallax Quality (n= 194)
$\sigma_\varpi = 0.2$ mas
Parallax Error ($\sigma_\varpi$) [mas]
Parallax ($\varpi$) [mas]
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
2
4
6
8
10
12
14


APOGEE DR17 Field: [K2_C10_281+55] Stellar Population Parameters (n= 194)
logg_gspphot
2.5
3.0
3.5
4.0
4.5
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
Metallicity ([M/H]) (gspphot)
9000
8000
7000
6000
5000
4000
3000
Effective Temperature ($T_{\rm eff}$) [K]

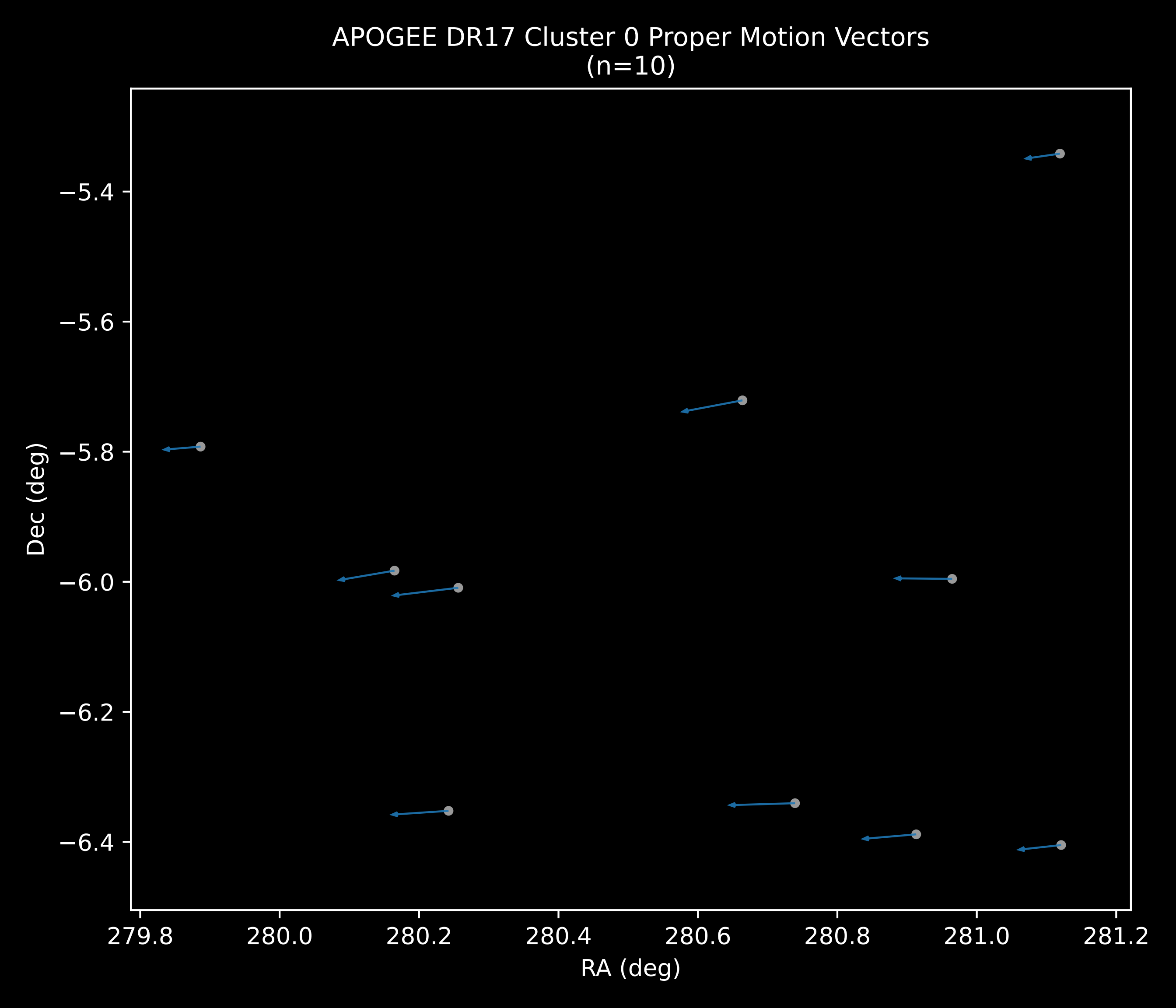


GAPOGEE DR17 Field [K2_C10_281+55] RUWE Distribution (n= 188)
RUWE = 1.4 Quality Cut
Star Count
0
20
40
60
80
1.0
1.5
2.0
2.5
3.0
3.5
4.0
RUWE (Renormalised Unit Weight Error)

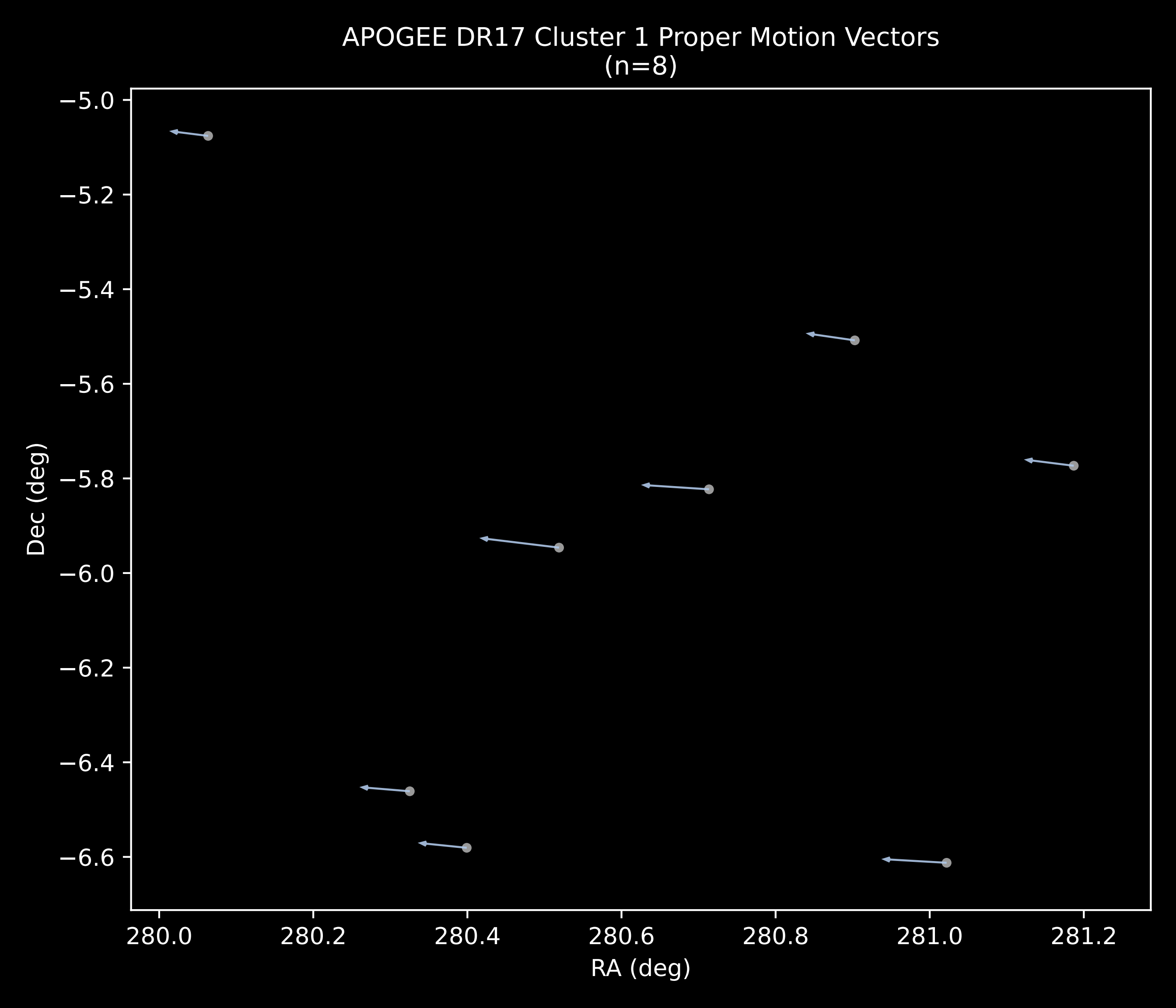

APOGEE DR17 Field: [K2_C10_281+55]
Proper Motion Vectors for Gaia Star Field (n= 194)
Stars
Dec (deg)
RA (deg)
−5.00
−5.25
−5.50
−5.75
−6.00
−6.25
−6.50
−6.75
279.75
280.00
280.25
280.50
280.75
281.00
281.25
281.50



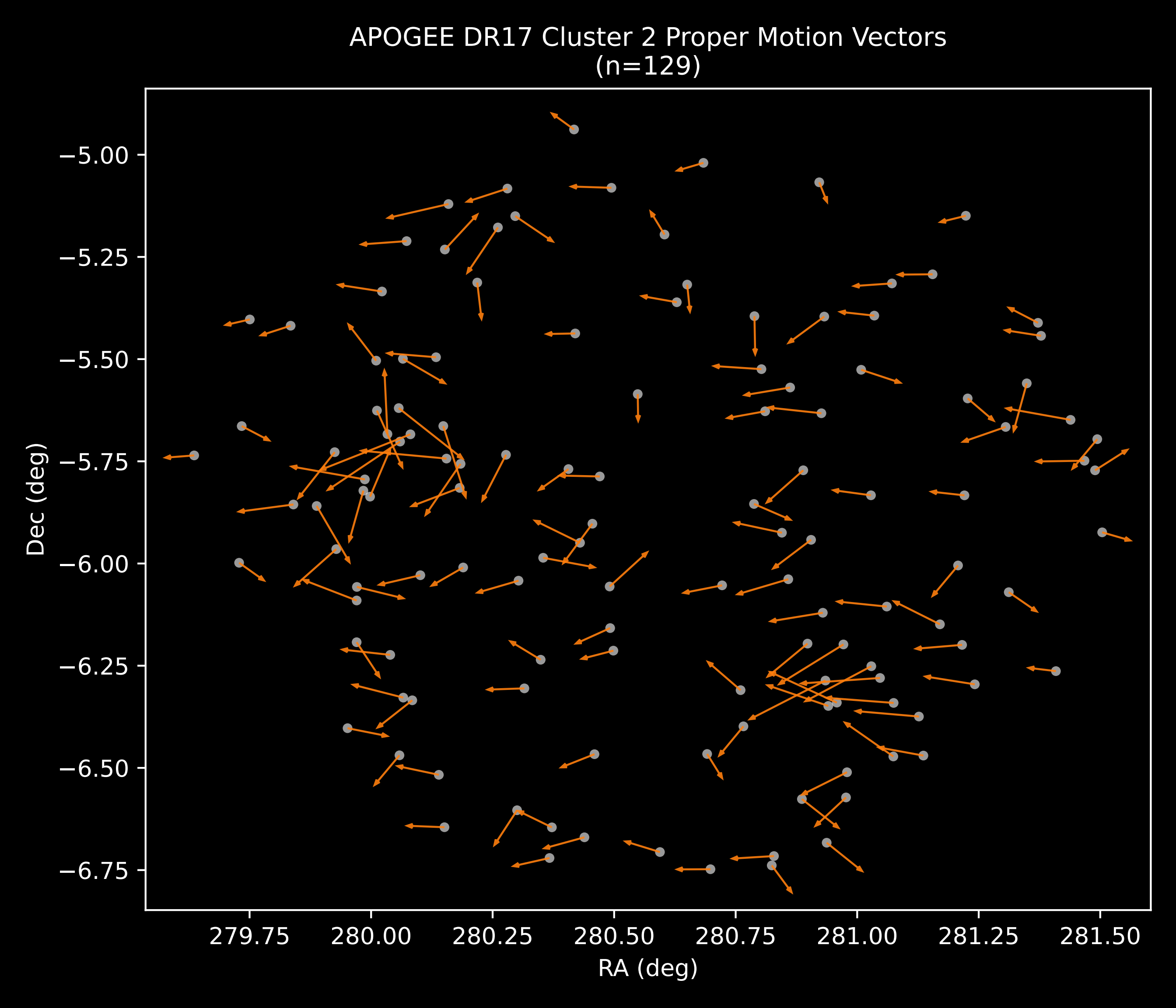
APOGEE DR17 Field [K2_C10_281+55]
Proper Motion Vectors with HDBSCAN
(n=194)
HDBSCAN Clusters
Stars
Cluster 0 (n=10)
Cluster 1 (n=8)
Cluster 2 (n=129)
Noise (n=47)
Dec (deg)
RA (deg)
−5.00
−5.25
−5.50
−5.75
−6.00
−6.25
−6.50
−6.75
279.75
280.00
280.25
280.50
280.75
281.00
281.25
281.50



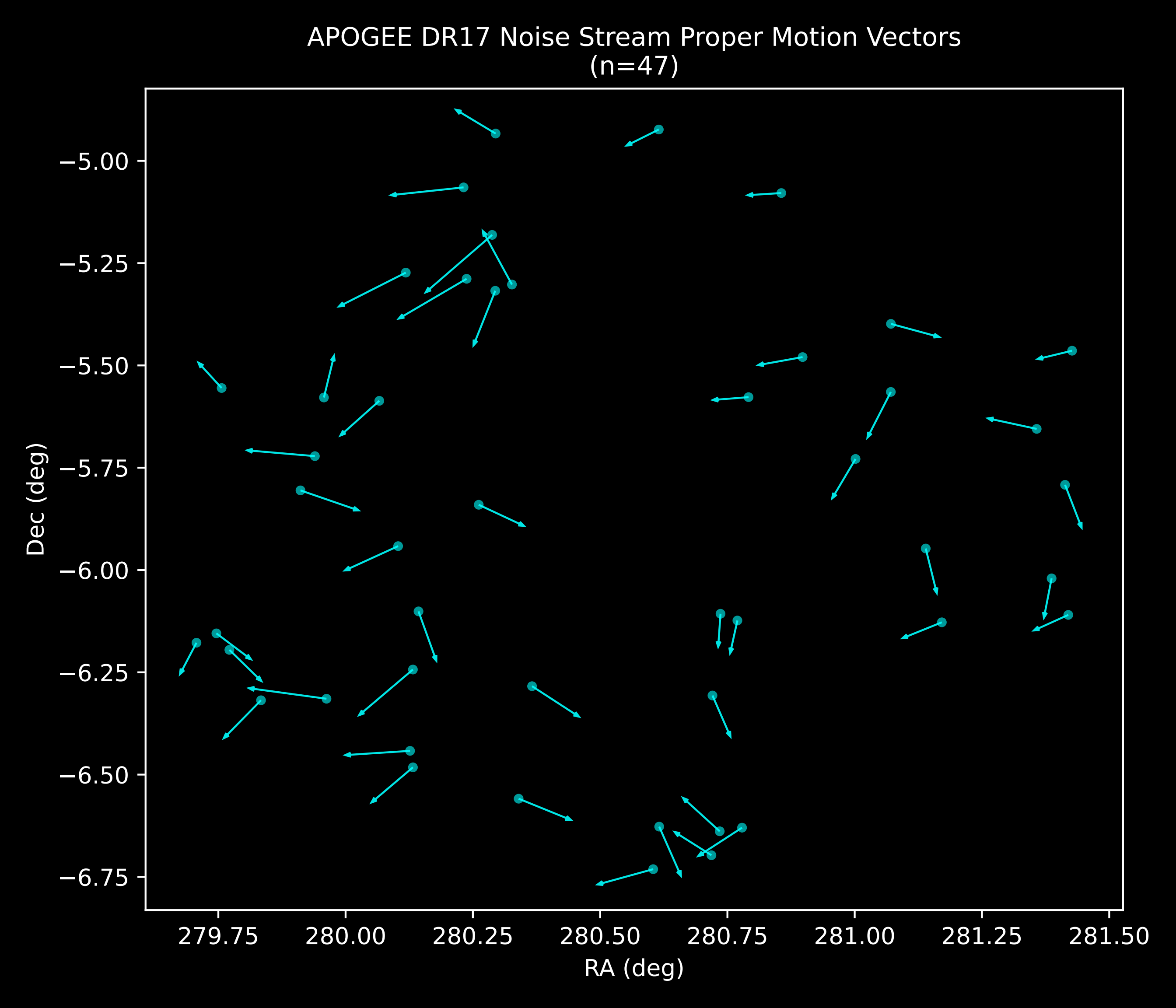
APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=10)
−5.4
−5.6
−5.8
−6.0
−6.2
−6.4
Dec (deg)
279.8
280.0
280.2
280.4
280.6
280.8
281.0
281.2
RA (deg)



APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=8)
Dec (deg)
−5.0
−5.2
−5.4
−5.6
−5.8
−6.0
−6.2
−6.4
−6.6
RA (deg)
280.0
280.2
280.4
280.6
280.8
281.0
281.2



APOGEE DR17 Cluster 2 Proper Motion Vectors
(n=129)
−5.00
−5.25
−5.50
−5.75
−6.00
−6.25
−6.50
−6.75
Dec (deg)
279.75
280.00
280.25
280.50
280.75
281.00
281.25
281.50
RA (deg)



APOGEE DR17 Noise Stream Proper Motion Vectors
(n=47)
Dec (deg)
RA (deg)
−5.00
−5.25
−5.50
−5.75
−6.00
−6.25
−6.50
−6.75
279.75
280.00
280.25
280.50
280.75
281.00
281.25
281.50